

Lesson 20 Functions

1. For each of the following relation, determine if it is a function. If it is not, explain why not. If it is, give its domain and image, and find $f(2)$.
 - a) $R = \{(1,2), (2,1)\}$
 - b) $R = \{(2,2), (2,3)\}$
 - c) $R = \{(2,2), (3,2)\}$
 - d) $R = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} : y = 2x\}$
 - e) $R = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} : xy = 0\}$
 - f) $R = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x + y = 1\}$
 - g) $R = \{(x,y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 1\}$
2. Let $A = \{-2, -1, 0, 1, 2\}$. Consider the function $f: A \rightarrow \mathbb{Z}$, $f(x) = (x + 1)^2$.
 - a) Find the image of f .
 - b) Find $f(\{0, 1, 2\})$.
 - c) Find $f^{-1}(\{0, 1\})$.
 - d) Let $S = \{x \in A : |x| > 1\}$. Find the image of S under f .
 - e) Let $T = \{x \in \mathbb{Z} : x < 5\}$. Find the inverse image of T under f .
3. Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Consider the function $f: \mathbb{N} \rightarrow A$, $f(x) =$ the ones digit of x . For example, $f(12) = 2$, $f(30) = 0$, etc.
 - a) Let $S = \{x \in \mathbb{N} : 5|x\}$. Find $f(S)$.
 - b) Let $T = \{0\}$. Find $f^{-1}(T)$.
4. Let A and B be sets and $f: A \rightarrow B$. Let $R, S \subseteq A$ and $T, V \subseteq B$. Prove that:
 - a) $f(R \cup S) = f(R) \cup f(S)$
 - b) $f^{-1}(T \cup V) = f^{-1}(T) \cup f^{-1}(V)$
 - c) $f^{-1}(T \cap V) = f^{-1}(T) \cap f^{-1}(V)$

The following problems are optional.

5. Let A and B be sets and $f: A \rightarrow B$. Let $R, S \subseteq A$ and $T, V \subseteq B$.:
 - a) Prove that $S \subseteq f^{-1}(f(S))$ but $f^{-1}(f(S)) \neq S$
 - b) Prove that $f(f^{-1}(T)) \subseteq T$ but $f(f^{-1}(T)) \neq T$
6. Let $f: A \rightarrow B$. Define a relation R on A : xRy if and only if $f(x) = f(y)$. Prove that R is an equivalence relation.